# Autodeformalization of Cramér-Rao Bounds: General, CMOS, and Event Sensors

## 1 Introduction

This document formally deformalizes the Lean 4 proofs corresponding to three distinct measurement regimes: the generalized Cramér-Rao Lower Bound (CRLB), its application to standard CMOS integration, and its modification for the differential log-intensity framework of event-based sensors.

## 2 General Cramér-Rao Lower Bound

**Definition 1** (Parameter Space and Fisher Information)**.** *Let $\iota$ be a finite index set such that the parameter space is $\Theta = \mathbb{R}^{|\iota|}$. Given raw sensor data $x \in Raw$, a forward model $g(\theta, x)$, and noise variance $\sigma(x)^2$, the Fisher Information Matrix $F(x) \in \mathbb{R}^{|\iota| \times |\iota|}$ for a single measurement is defined as:*

$$F(x)_{ij} = \frac{\frac{\partial g(\theta,x)}{\partial \theta_i} \frac{\partial g(\theta,x)}{\partial \theta_j}}{g(\theta, x) + \sigma(x)^2} \tag{1}$$

*The total information matrix for a field of view (FOV) is $F_{image}(\theta) = \sum_{x \in FOV} F(x)$.*

**Definition 2** (Estimator and Covariance)**.** *An estimator $d : (Raw \to \mathbb{R}) \to \Theta$ is unbiased under measure $\mu$ if its expected value is the true parameter $\theta$. Its covariance matrix is $C = \mathbb{E}_\mu[(d(m) - \theta)(d(m) - \theta)^T]$.*

**Theorem 1** (Cramér-Rao Inequality)**.** *For any unbiased estimator $d$ with covariance matrix $C$ and an invertible total Fisher Information Matrix $F$, the matrix difference $C - F^{-1}$ is positive semi-definite (i.e., $C \succeq F^{-1}$).*

*Proof.* In the formal specification (`cramer_rao_inequality`), this is admitted via the `sorry` tactic. We deformalize this by defining the standard regularity conditions required for probability distributions.

**Assumption 1.** *We assume the operations of integration (expectation) and differentiation with respect to $\theta$ can be interchanged. We additionally assume the score function has an expected value of zero.*

Under Assumption 1, the covariance between the estimator error and the score function analytically evaluates to the identity matrix. Applying the Cauchy-Schwarz inequality or the Schur complement to the block covariance matrix of the estimator and the score yields $C - F^{-1} \succeq 0$, completing the logical proof admitted by the `sorry` block. □

## 3 CMOS Sensor Model and Asymptotics

For a CMOS sensor, the expected photoelectron count is modeled as $g_m(\theta) = N s_m(\alpha, x_c, y_c, z_c) + b$. We define a scalar standard error $SE(T)$ parametrized by integration time $T$:

$$SE(T) = \sqrt{\frac{\Phi T s_m + \beta T + \sigma^2}{\Phi^2 T^2 s_m^2}} \tag{2}$$

**Lemma 1** (Photon-Noise Limit)**.** *Assuming $\Phi > 0$, $s_m > 0$, and $\beta \geq 0$, as $T \to \infty$, $SE(T) \in \Theta(1/\sqrt{T})$.*

*Proof.* In `std_error_photon_limit`, the formal proof delegates the limit evaluation to `sorry`.

**Assumption 2.** *We algebraically assume that* $\lim_{T\to\infty} \frac{\sigma^2}{T} = 0$*, causing the asymptotic ratio* $\frac{1/\sqrt{T}}{SE(T)}$ *to rigorously converge to the strictly positive constant* $\sqrt{\frac{\Phi^2 s_m^2}{\Phi s_m + \beta}}$*.*

This validates the bound, demonstrating that read noise becomes negligible over long integrations compared to Poisson counting statistics. □

**Lemma 2** (Read-Noise Limit)**.** *Assuming* $\Phi > 0, s_m > 0, \sigma^2 > 0$*, as* $T \to 0^+$*,* $SE(T) \in \Theta(1/T)$*.*

*Proof.* In `std_error_read_noise_limit`, the formal limit computation is delegated to `sorry`.

**Assumption 3.** *We assume the terms bound by* $T$ *in the numerator vanish, causing the limit ratio* $\frac{1/T}{SE(T)}$ *to converge to* $\sqrt{\frac{\Phi^2 s_m^2}{\sigma^2}}$*.*

This verifies the $\Theta(1/T)$ bound, confirming that for vanishingly small integration intervals, the fixed readout variance $\sigma^2$ strictly dominates. □

# 4 Event Sensor Model and Asymptotics

An event sensor operates by evaluating the differential log-intensity between frames. The measurement is approximated as a Normal distribution derived via the Delta method for the ratio of two Poisson variables $\mu(T) = \Phi_\mu T + \beta T$ and $\nu(T) = \Phi_\nu T + \beta T$. The mean and variance are approximated as $m = \frac{\nu}{\mu}$ and $v = \frac{\nu}{\mu^2} + \frac{\nu^2}{\mu^3}$. The total scalar Fisher Information $I(T)$ scales inversely with $v$, and the standard error is given by $e(T) = 1/\sqrt{I(T)}$.

**Lemma 3** (Event Photon-Noise Limit)**.** *For* $\Phi_\mu, \Phi_\nu > 0$*,* $\beta \geq 0$*, and non-zero parameter gradients, as* $T \to \infty$*,* $e(T) \in \Theta(1/\sqrt{T})$*.*

*Proof.* In `event_std_error_photon_limit`, this is admitted via `sorry`.

**Assumption 4.** *We assume the limit of the expanded Fisher Information scales exactly proportionally to* $T$*. Specifically, as* $T \to \infty$*, the variance* $v \sim O(T^{-1})$*, making* $1/v \sim O(T)$*. Given* $(\nabla m)^2 \sim O(1)$*, the primary information coefficient* $\frac{1}{v}(\nabla m)^2 \sim O(T)$*. The secondary variance derivative coefficient* $\frac{1}{2v^2}(\nabla v)^2$ *scales as* $O(T^2) \cdot O(T^{-2}) \sim O(1)$ *and becomes asymptotically negligible.*

This limits the information bound to $I(T) \sim O(T)$, resolving the standard error scalar to $e(T) = I(T)^{-1/2} \sim \Theta(1/\sqrt{T})$. □

**Lemma 4** (Event Read-Noise Limit)**.** *For* $\Phi_\mu, \Phi_\nu > 0, \beta > 0$*, as* $T \to 0^+$*,* $e(T) \in \Theta(1/T)$*.*

*Proof.* In `event_std_error_read_noise_limit`, this is admitted via `sorry`.

**Assumption 5.** *As* $T \to 0^+$*, the variance* $v(\mu, \nu) \sim O(T^{-1})$ *due to the linear decay of signal photons yielding to constant baseline threshold noise and quantization errors. Consequently, the squared parameter gradients scale such that the total scalar Fisher Information limits tightly to* $O(T^2)$*.*

By the limit comparison test, $e(T) = I(T)^{-1/2} \sim \Theta(1/T)$. □

# 5 References


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